Identifiability and Convexification of a Band-Limited Unassigned Distance-Geometry Problem

Let $X = (x_1, \ldots, x_N) \in (\mathbb{R}^3)^N$ carry labels $e_i \in E$ and weights $b_i > 0$. Quotient by rigid motions (translations and rotations, but *not* reflections): $\mathcal{X} = (\mathbb{R}^3)^N / E(3)$. The weighted pair-distance measure and its measurement are

$$\rho_X = \sum_{i<j} w_{ij}\, \delta_{\|x_i - x_j\|}, \quad w_{ij} = b_i b_j \geq 0, \qquad F(X) = R_{Q_{\max},\sigma}\, \rho_X \in Y,$$

where $R_{Q_{\max},\sigma}$ is a known compact smoothing operator: a band-limit $Q_{\max}$ (real-space resolution $\Delta r \approx \pi / Q_{\max}$) followed by convolution of width $\sigma > 0$. Its *information capacity*

$$m_{\text{eff}}(Q_{\max}, \sigma) = \dim \operatorname{span}\{R_{Q_{\max},\sigma} \delta_t : t \in [0, R_{\max}]\} \approx \lceil R_{\max} / \Delta r \rceil$$

is finite, with $m_{\text{eff}} \to \infty$ as $Q_{\max} \to \infty$ (exact-distance limit) and $m_{\text{eff}} \to 0$ as $Q_{\max} \to 0$, and $\operatorname{rank}(DF_X) \leq \min(d_C, m_{\text{eff}})$. A *prior* is a closed $C \subset \mathcal{X}$, a smooth manifold of dimension d_C realizable in internal coordinates (a graph of fixed edge lengths and angles with d_C free dihedrals). F is an instance of the unassigned distance geometry problem [3]; in the motivating application C encodes chemical constraints, which play no formal role below.

Known facts.

1. For $R = \text{id}$, injectivity of ρ up to rigid motion is the global-rigidity problem; in 3D generic global rigidity requires redundant rigidity and stressability and is subtler than edge-counting, and special configurations are generically *not* globally rigid (flip ambiguities persist).

2. In 1D (turnpike/beltway) with unassigned noisy distances, projected gradient descent with a spectral initializer converges locally linearly to the global optimizer under explicit conditions [1]; for integer sets, sparsity yields polynomial-time recovery up to shift and reversal w.h.p. [2].

3. A configuration and its mirror image have identical $F(X)$, so identifiability holds at best modulo $\mathbb{Z}_2$ (fiber $K \geq 2$ for chiral configurations).

Open problem. For $y = F(X_\star)$ and $\mathcal{L}_y(X) = \|F(X) - y\|_Y^2|_C$:

(P1) *Identifiability:* $F|_C$ injective modulo reflection (fibers of size 1, or 2 = the mirror pair).

(P2) *Local identifiability:* the Fisher operator $I_X = (DF_X)^*(DF_X)$ has full rank on $T_X C$ for all $X \in C$.

(P3) *ε-convexity:* $\mathcal{L}_y|_C$ is geodesically ε-convex on the component of $X_\star$ with a unique stationary point (no spurious local minima).

Problem 1. *Characterize the priors C, as a function of m_{eff}, for which (P1) $\Leftrightarrow$ (P2) $\Leftrightarrow$ (P3) hold for generic $X_\star \in C$, and find the* minimal *such prior. In particular:*

(a) *Prove or disprove a sharp threshold at* $m_{\text{eff}} = d_C$ *(modulo reflection), with counterexamples at* $m_{\text{eff}} = d_C - 1$.

(b) *Find the maximal admissible* d_C *(least external information) and the trade-off curve* $d_C^*(Q_{\max})$.

(c) *Show the threshold coincides with vanishing of* $I_{\text{deficit}} = h(X \mid C) - I(F(X); X \mid C)$ *(up to* $\log 2$*).*

This is generic-global-rigidity theory for band-limited, weighted, soft data on a constraint manifold — the regime open at the intersection of [1, 2, 3].

References

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